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Energy Consumption, Economic Growth, and CO₂ Emissions in GCC Countries: Panel Evidence and the Environmental Kuznets Curve

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Abstract

The Gulf Cooperation Council (GCC) countries consistently rank among the highest per capita CO₂ emitters globally, yet rigorous empirical analysis of the structural drivers of these emissions in the post-Paris Agreement era remains scarce. This study investigates the determinants of CO₂ emissions per capita across six GCC economies—Bahrain, Kuwait, Oman, Qatar, Saudi Arabia, and the United Arab Emirates—over the period 2015–2022, using pooled ordinary least squares (OLSs) and country fixed effects (FEs) panel regression models with country-clustered standard errors. The focal explanatory variable is energy use per capita, complemented by GDP per capita, trade openness, urbanization, foreign direct investment (FDI), and industry value added as controls. A quadratic income term explicitly tests the environmental Kuznets curve (EKC) hypothesis. Results consistently show that energy use is the dominant driver of emissions. The EKC hypothesis is supported in the FE framework. The implied turning point of approximately USD 85,500 per capita (constant 2015 USD) is already exceeded by Qatar (panel mean: USD 114,835) and approached by the UAE (USD 71,434), while Bahrain (USD 55,681), Kuwait (USD 51,531), Saudi Arabia (USD 61,232), and Oman (USD 38,591) remain on the EKC's rising slope, consistent with their continued emissions' growth trajectories. Urbanization exerts a significant positive within-country effect on emissions. Trade openness reduces emissions in cross-sectional specifications, while FDI is systematically insignificant. These findings support energy efficiency reforms, renewable energy expansion, and low-carbon urban planning as the most effective policy levers for GCC decarbonization.

Keywords: CO₂ emissions; energy consumption; environmental Kuznets curve; GCC countries; panel data; fixed effects; sustainability; STIRPAT



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1. Introduction

The challenge of reconciling economic growth with environmental sustainability is particularly acute for the Gulf Cooperation Council (GCC)—a block of six high-income, hydrocarbon-dependent economies whose structural reliance on fossil fuels generates some of the highest per capita CO₂ emissions in the world. Qatar recorded over 41 metric tons of CO₂ per capita in 2024, while Bahrain, Kuwait, Oman, Saudi Arabia, and the United Arab Emirates (UAEs) each exceeded 20 metric tons per capita annually. Together, the GCC contributes approximately 1.5 billion tons of CO₂ per year—roughly 4% of global

emissions—while hosting some of the world’s largest known hydrocarbon reserves [1]. GCC consumption-based CO₂ emissions nearly doubled from 540.79 MtCO₂ in 2003 to 1090.93 MtCO₂ in 2020 [2]. Despite ambitious Vision 2030 agendas and net-zero pledges—the UAE targeting net zero by 2050, and Saudi Arabia and Kuwait by 2060—the total budgetary cost of fossil fuel subsidies has increased in recent years, driven primarily by the global energy price surge of 2021–2022 rather than deliberate rate increases; nonetheless, the persistence of implicit subsidies—through below-cost energy pricing—continues to dampen incentives for energy efficiency and renewable adoption [2].

The economics literature consistently identifies energy consumption, income growth, trade openness, urbanization, and FDI as core structural determinants of CO₂ emissions [3–5]. However, most GCC-specific empirical work predates the post-Paris Agreement era, leaving the 2015–2022 period—characterized by Vision 2030 structural reforms, the COVID-19 energy demand shock, and accelerating renewable energy investment—largely unexamined. Furthermore, the environmental Kuznets curve (EKC) hypothesis—positing an inverted-U relationship between per capita income and environmental degradation—remains contested in the GCC context, with divergent findings across studies using different methodologies and time periods [6,7].

This paper fills three specific gaps. First, it provides the most up-to-date panel analysis of GCC emission determinants, with descriptive coverage extending to 2024 using the most recently published Global Carbon Budget data, and a regression sample spanning 2015–2022—the period for which all key variables, including the focal energy use per capita series, are jointly available. This regression window is itself more recent than the large majority of existing GCC panel studies, which end no later than 2018 or 2020. Second, it employs country fixed effect estimators with clustered standard errors, isolating within-country dynamics obscured by cross-sectional approaches. Third, it provides an explicit EKC test through a quadratic income specification, contributing to the contested GCC EKC debate.

The remainder of the paper is structured as follows. Section 2 reviews the literature and develops hypotheses. Section 3 describes data and methodology. Section 4 presents empirical results. Section 5 discusses findings. Section 6 concludes.

2. Literature Review and Hypotheses Development

2.1. Theoretical Framework

Three complementary frameworks underpin this study. The *environmental Kuznets curve* (EKC) hypothesis, originating from Grossman and Krueger [8] and formalized by Panayotou [9], posits an inverted-U relationship between per capita income and environmental degradation—a relationship extensively surveyed by Dinda [10]. At low-income levels, the *scale effect* of economic expansion raises emissions. Beyond an income turning point, the *composition effect* (structural shift toward cleaner industries) and the *technique effect* (adoption of cleaner technologies) cause emissions to decline [10].

The *STIRPAT* (Stochastic Impacts by Regression on Population, Affluence and Technology) framework [11] decomposes environmental impact into population, affluence, and technology components, permitting independent econometric testing of each driver. The *scale–composition–technique* (SCT) decomposition of Antweiler et al. [10] extends this to trade and FDI as follows: each operates through scale (more activity), composition (sector reallocation), or technique (technology diffusion) channels. Together, these frameworks structure the model specification and interpretation of results throughout the paper.

2.2. Energy Consumption and CO₂ Emissions

Energy consumption is the most consistently identified proximate driver of CO₂ emissions, but the magnitude of the energy elasticity varies substantially across methodologies and country groups. For the GCC and MENA region, Harrathi and Almohaimeed [6] use Pooled Mean Group (PMG) estimation over 1980–2018, reporting long-run energy elasticities exceeding unity for the GCC sub-panel; Arouri et al. [12] and Asif et al. [13] confirm the positive energy–emission relationship through panel cointegration. These studies share the following common limitation: they treat the GCC as a structurally homogeneous unit over long spans that predate the post-Paris reform era, masking the within-country dynamics most relevant for near-term policy. The global evidence corroborates the sign but underscores specification sensitivity as follows: Balogh and Jámor [4] show in a 168-country dynamic GMM panel that coal-based electricity raises emissions while renewables reduce them—a distinction obscured by total energy aggregates. Banday and Aneja [14] document bidirectional causality in G7 countries, flagging the simultaneity concern that also applies to the present study. The country-specific nature of the energy–emission relationship is further illustrated by Ang [15], who identifies long-run cointegration between energy use and CO₂ within a single-country setting, underscoring that within-country identification is feasible and policy-relevant. Existing GCC studies rely on long-run aggregated approaches that predate the post-Paris reform period, leaving the within-country energy–emission dynamics of 2015–2022 unexamined.

2.3. The EKC Hypothesis: Global and GCC-Specific Evidence

The EKC literature produces contradictory results because the following three methodological choices jointly determine the estimated turning point: the estimator (pooled cointegration vs. within-country fixed effects), whether energy consumption is controlled separately from the income quadratic, and the income range covered by the sample. The canonical BMA analysis of Aller et al. [3] confirms GDP per capita as a robust positive predictor of CO₂, consistent with a dominant scale effect, while Balogh and Jámor [4] confirm the EKC in a dynamic GMM global panel. Bae et al. [16] find only a linear income–emission relationship for post-Soviet countries, underscoring that the EKC is not universal but depends on institutional context and income range. For the GCC, this methodological divide maps onto a temporal one as follows: Harrathi and Almohaimeed [6] find no EKC for oil-rich GCC panels using data ending in 2018, whereas Adow et al. [7] and Toukabri and Chaouachi [17]—drawing on longer or more recent samples with cross-sectional dependence-robust and FMOLS/DOLS estimators—confirm the EKC for all six GCC economies.

The divergence between earlier GCC studies and more recent findings motivates an explicit EKC test using country fixed effects, with the income quadratic specified jointly with energy use.

2.4. Urbanization and CO₂ Emissions

Urbanization’s environmental effect is theoretically ambiguous. The agglomeration channel predicts lower per-unit energy demand through shared infrastructure and compact land use [5]; the energy-demand amplification channel—dominant in hot, arid regions—predicts higher per capita emissions as expanding urban footprints increase demand for cooling, desalination, and car-based mobility [13]. The GCC literature consistently finds the amplification channel dominant: Harrathi and Almohaimeed [6] and Asif et al. [13] both report positive urbanization–CO₂ relationships for GCC economies, consistent with global emerging-economy evidence from Kwakwa [18] and Ali et al. [19]. What distinguishes the GCC is structural intensity: with urban population shares averaging 91% in our sample,

incremental urbanization occurs at the high-density urban fringe where per-unit energy demand for cooling and transport is greatest. No existing GCC study isolates this within-country urbanization effect during the post-2015 period, motivating H5.

2.5. Trade Openness, FDI, and Industrialization

Trade openness and FDI present competing theoretical predictions resolved only by country context. For trade, the SCT decomposition [11] predicts that high-income economies—where technique and composition effects dominate—exhibit negative trade–emission relationships; Aller et al. [3] confirm this in BMA analysis, and Harrathi and Almohaimeed [6] confirm the significant negative long-run effect for GCC economies. However, in hydrocarbon-exporting economies, measured trade openness (exports plus imports as a share of GDP) is procyclically linked to oil prices, because export revenues and nominal GDP co-move. Year-to-year changes in this ratio reflect commodity price cycles rather than genuine shifts in trade integration—a Dutch Disease dynamic that dilutes within-country identifying variation and explains why the negative trade coefficient, robust in cross-sectional specifications, typically loses significance in fixed effect models.

For FDI, the pollution haven hypothesis [16]—predicting dirty-sector relocation toward laxly regulated hosts—competes with the pollution halo hypothesis, which holds that multinational investment brings cleaner technologies [3]. GCC evidence, including Harrathi and Almohaimeed [6], consistently finds FDI statistically insignificant, suggesting that neither channel dominates in a region where FDI concentrates on financial services and construction rather than emission-intensive manufacturing. Caglar et al. [20] confirm for BRICS that natural resource dependence worsens environmental quality, consistent with the GCC’s hydrocarbon-export structure. The relative importance of these channels in the GCC’s post-2015 reform context remains unresolved, motivating H4.

2.6. Research Gap and Hypotheses

Three gaps motivate this study. First, existing GCC panel studies end before 2018 [6,12], missing the post-Paris Agreement era of Vision 2030 structural reforms. Second, newer studies [7,17] use long-run cointegration methods that do not exploit within-country time variation as precisely as fixed effects with clustered standard errors. Third, no study combines a quadratic EKC test with country fixed effects for the recent GCC panel. The following hypotheses are proposed:

H1. *Energy consumption per capita is positively and significantly associated with CO₂ per capita in GCC countries.*

H2. *GDP per capita exerts a positive (scale) effect on CO₂ emissions.*

H3. *The EKC hypothesis holds: the quadratic GDP per capita term is negative and significant.*

H4. *Trade openness is negatively associated with CO₂ through the technique effect of international market integration.*

H5. *Urbanization raises CO₂ within GCC countries, reflecting energy-intensive urban expansion in desert environments.*

3. Data and Methodology

3.1. Data Sources and Sample

The study uses a balanced panel of six GCC countries—Bahrain, Kuwait, Oman, Qatar, Saudi Arabia, and the UAE—over the period 2015–2022. This period encompasses the

post-Paris Agreement era, the launch and early implementation of Vision 2030 strategies, the COVID-19 energy demand shock of 2020, and the subsequent economic recovery. Data are sourced from the World Bank World Development Indicators (WDIs) [21] and the International Energy Agency (IEA) [22]. The panel yields a maximum of 60 country-year observations for variables available through 2024 (e.g., CO₂, GDP per capita, urbanization, and FDI); the energy use per capita variable is available only through 2022, yielding 48 observations for the main regression sample. Descriptive statistics covering the full 2015–2024 panel are reported in Tables 2 and 3.

The dependent variable is CO₂ emissions per capita (metric tons). The focal independent variable is energy use per capita (kg of oil equivalent). Controls include: GDP per capita (constant 2015 USD); trade openness (exports plus imports as a percentage of GDP); urban population share (%); FDI net inflows (% of GDP); and industry value added (% of GDP) as a robustness extension. All continuous variables are natural-log-transformed except FDI. FDI net inflows are retained in levels—expressed as a percentage of GDP—because the variable contains negative values arising when net outflows exceed inflows, and because it is already a dimensionless ratio that does not require scale normalization (Harrathi and Almohaimeed [6]; Aller et al. [3]). An alternative robustness specification substitutes electric power consumption (kWh per capita) for total energy use.

For the four variables available through 2024—CO₂ per capita, GDP per capita, urbanization, and FDI—we compare the panel means over 2015–2022 (the regression sample) with the extended period 2015–2024 (the descriptive sample). The differences are minor: mean CO₂ per capita changes from 23.9 to 24.1 metric tons; mean GDP per capita from USD 65,112 to USD 65,551; and mean urban population from 91.1% to 91.2%. FDI exhibits somewhat greater inter-annual variation but no structural break. Pairwise correlations among these four variables are virtually identical across the two sub-periods (maximum correlation difference: 0.04).

Table 1 provides a complete summary of all variable definitions, measurement units, and data sources used in the analysis.

Table 1. Variable definitions and data sources.

Variable	Definition	Unit	Source
lnCO ₂	Log CO ₂ emissions per capita (dependent)	metric tons pc	IEA [22]
lnEnergy	Log energy use per capita (focal variable)	kg oil eq. pc	WDI [21]
lnGDPpc	Log GDP per capita, constant 2015 USD	USD	WDI [21]
(lnGDPpc) ²	Squared lnGDPpc—EKC test term	(dimensionless)	Authors
lnTrade	Log trade openness (exports + imports/GDP)	% of GDP	WDI [21]
lnUrban	Log urban population share	% of total	WDI [21]
FDI	FDI net inflows (% of GDP)	% of GDP	WDI [21]
lnIndustry	Log industry value added (% of GDP)	% of GDP	WDI [21]
lnElectric	Log electric power consumption pc (robustness)	kWh pc	WDI [21]

Note: pc = per capita. All log transformations are natural logarithms. Period: 2015–2024 for descriptive analysis; regression sample constrained to 2015–2022 by energy use data availability (N = 48).

3.2. Econometric Specification

Following the STIRPAT framework [23] augmented with trade, FDI, and urbanization controls standard in the GCC emissions literature [6], the pooled OLS baseline is:

$$\ln CO_2 it = \alpha + \beta_1 \ln Energy it + \beta_2 \ln GDPpc it + \beta_3 \ln Trade it + \beta_4 \ln Urban it + \beta_5 FDI it + \varepsilon it \quad (1)$$

The preferred country fixed effect (FE) specification introduces country-specific intercepts α_i to control for time-invariant unobserved heterogeneity in hydrocarbon endowments, energy subsidy regimes, and institutional frameworks:

$$\ln\text{CO}_2\text{ it} = \alpha i + \beta_1 \ln\text{Energy it} + \beta_2 \ln\text{GDPpc it} + \beta_3 \ln\text{Trade it} + \beta_4 \ln\text{Urban it} + \beta_5 \text{FDI it} + \varepsilon\text{ it} \quad (2)$$

To test H3, Model 3 augments Equation (2) with a quadratic income term:

$$\ln\text{CO}_2\text{ it} = \alpha i + \beta_1 \ln\text{Energy it} + \beta_2 \ln\text{GDPpc it} + \beta_3 (\ln\text{GDPpc})^2\text{ it} + \beta_4 \ln\text{Trade it} + \beta_5 \ln\text{Urban it} + \beta_6 \text{FDI it} + \varepsilon\text{ it} \quad (3)$$

EKC support requires $\beta_2 > 0$ and $\beta_3 < 0$. Models 4 and 5 add industry value added and substitute electric power consumption, respectively, as robustness checks. Standard errors in all FE specifications are clustered at the country level to address within-panel serial correlation [24]. Multicollinearity is assessed via Variance Inflation Factors (VIFs) for the pooled OLS specification. All VIF values fall below 5 ($\ln\text{Energy} = 4.12$; $\ln\text{GDPpc} = 4.38$; $\ln\text{Trade} = 2.17$; $\ln\text{Urban} = 3.86$; $\text{FDI} = 1.74$; and mean $\text{VIF} = 3.25$), indicating multicollinearity is not a concern. The fixed-effect estimator further mitigates collinearity by exploiting within-country time variation. Cross-sectional dependence is tested using the Pesaran (2004) [25] CD test. The statistic of $\text{CD} = 1.68$ ($p = 0.093$) fails to reject cross-sectional independence at the 5% level, supporting the FE estimator with country-clustered standard errors as the preferred specification.

A limitation of the pooled OLS and fixed-effect estimators is their susceptibility to simultaneity bias if energy consumption or income are jointly determined with CO_2 emissions. The fixed-effect estimator partially mitigates omitted variable bias by removing time-invariant country-level confounders, but it does not address time-varying simultaneity. Valid instruments for energy use in a short GCC panel are difficult to identify, and dynamic GMM is unreliable at $N = 6$. The estimates are therefore best interpreted as conditional panel associations rather than identified causal effects, and causal identification via instrumental variables or natural experiments is recommended for future research.

4. Results

4.1. Descriptive Statistics

Table 2 reports descriptive statistics for the full GCC panel over 2015–2024. Mean CO_2 per capita is 24.1 metric tons—more than five times the global average of approximately 4.7 metric tons over the same period. Qatar records the highest country mean at 37.9 metric tons, reflecting its large LNG processing sector and comparatively small national population. Oman records the lowest at 15.8 metric tons. Energy use per capita averages 9622 kg of oil equivalent, with Qatar leading at over 16,000 kg. GDP per capita averages USD 65,551, ranging from Oman (USD 38,591) to Qatar (USD 114,835). Trade openness averages 109.5% of GDP, reflecting the highly open nature of GCC economies. Urban population averages 91.2%, with Bahrain, Kuwait, and Qatar fully urbanized. FDI averages 2.6% of GDP with considerable variability. Table 3 presents country-level means over 2015–2024.

Table 2. Descriptive statistics (GCC panel, 2015–2024).

Variable	Obs.	Mean	Std. Dev.	Min	Max
CO_2 per capita (metric tons)	60	24.105	6.954	14.622	41.271
Energy use (kg oil eq. pc)	48	9621.9	3316.6	5104.1	17,100.8
GDP per capita (USD)	60	65,551	24,572	36,721	123,071
Trade openness (% of GDP)	56	109.50	40.51	47.53	199.05
Urban population (%)	60	91.18	8.82	77.30	100.00
FDI net inflows (% of GDP)	60	2.581	3.480	−2.760	15.643
Electric power consumption (kWh pc)	49	14,274	4780	7987	22,689
Industry value added (% of GDP)	60	49.75	7.40	38.20	65.77

Note: pc = per capita. Period: 2015–2024 for all variables except energy use (2015–2022). Monetary values in constant 2015 USD.

Table 3. Country-level mean values (2015–2024).

Country	CO ₂ pc (t)	Energy (kg)	GDP pc (USD)	Trade (% GDP)	Urban (%)
Bahrain	23.83	10,231	55,681	146.8	100.0
Kuwait	23.92	9205	51,531	93.1	100.0
Oman	15.75	5687	38,591	95.9	79.1
Qatar	37.86	16,110	114,835	92.5	99.1
Saudi Arabia	21.21	7563	61,232	56.5	83.7
United Arab Emirates	22.06	8936	71,434	173.6	85.1

Note: means calculated over 2015–2024. Energy use values are means over 2015–2022. GDP per capita in constant 2015 USD. Source: authors' calculations.

4.2. Correlation Analysis

Table 4 presents the Pearson correlation matrix for the regression sub-sample (N = 48, 2015–2022). Log CO₂ per capita correlates very strongly with log energy use ($r = 0.972$), confirming the dominant role of fossil-fuel combustion in driving GCC emissions. The high correlation between lnEnergy and lnCO₂ ($r = 0.972$) and between lnGDPpc and lnCO₂ ($r = 0.909$) reflects the scale effects predicted by STIRPAT and EKC theory. Urbanization correlates positively with emissions ($r = 0.701$), consistent with Asif et al. [12]. FDI displays a negative raw correlation ($r = -0.490$). No pairwise correlation among regressors exceeds $|r| = 0.88$ in absolute value; the high lnEnergy–lnGDPpc correlation ($r = 0.878$) motivates the separate model specifications in Table 5. The within-country fixed-effect estimator mitigates cross-sectional multicollinearity by focusing on time-series variation.

Table 4. Pearson correlation matrix (regression sub-sample, 2015–2022, N = 48).

	lnCO ₂	lnEnergy	lnGDPpc	lnTrade	lnUrban	FDI
lnCO ₂	1.000					
lnEnergy	0.972	1.000				
lnGDPpc	0.909	0.878	1.000			
lnTrade	0.067	0.198	0.073	1.000		
lnUrban	0.701	0.769	0.425	0.209	1.000	
FDI	−0.490	−0.461	−0.389	0.358	−0.420	1.000

Note: all variables in natural logarithms except FDI. Source: authors' calculations.

Table 5. Panel regression results: determinants of lnCO₂ per capita in GCC countries (2015–2022).

Variable	Model 1 Pooled OLS	Model 2 FE Baseline	Model 3 FE + EKC	Model 4 FE + Industry	Model 5 FE Robustness
lnEnergy	0.580 *** (0.138)	0.722 *** (0.150)	0.738 *** (0.152)	0.705 *** (0.131)	—
lnElectric	—	—	—	—	0.280 (0.225)
lnGDPpc	0.204 ** (0.085)	0.195 (0.189)	8.839 *** (2.533)	0.232 * (0.126)	0.547 ** (0.211)
(lnGDPpc) ²	—	—	−0.389 *** (0.113)	—	—
lnTrade	−0.080 *** (0.021)	−0.007 (0.125)	−0.039 (0.109)	0.025 (0.090)	0.134 (0.166)
lnUrban	0.173 (0.257)	1.983 ** (0.942)	2.079 * (1.254)	2.075 ** (0.860)	−0.447 (2.142)
FDI	0.001 (0.003)	−0.001 (0.003)	−0.001 (0.005)	−0.001 (0.003)	−0.002 (0.002)
lnIndustry	—	—	—	−0.060 (0.099)	—
Constant	−4.803 *** (0.813)	—	—	—	—

Table 5. Cont.

Variable	Model 1 Pooled OLS	Model 2 FE Baseline	Model 3 FE + EKC	Model 4 FE + Industry	Model 5 FE Robustness
<i>Model Fit Statistics</i>					
Observations	48	48	48	48	48
Country FE	No	Yes	Yes	Yes	Yes
R ² (within/overall)	0.971	0.513	0.538	0.517	0.274

Notes: dependent variable: lnCO₂ per capita. Standard errors in parentheses: country-clustered for Models 2–5 (fixed effects); HC3-robust for Model 1 (pooled OLS). Significance levels: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Country fixed effects absorb the intercept in Models 2–5. Model 5 replaces total energy use with electric power consumption as a robustness check. A dash (—) indicates the variable is not included in that specification. R² reported as overall R² for Model 1 and within-country R² for Models 2–5; the two statistics are not directly comparable (see Section 4.3 for discussion). Regression sample: 2015–2022 (N = 48; six GCC countries, eight annual observations each).

4.3. Panel Regression Results

Table 5 presents results across five model specifications. Model 1 is the pooled OLS benchmark with HC3-robust standard errors. Models 2–5 employ country fixed effects with country-clustered standard errors. The regression sample covers 2015–2022 (N = 48) due to energy use data availability; the period 2015–2024 is used for all descriptive analyses.

Energy consumption (H1). Energy use per capita is positive, large, and significant at the 1% level across all specifications—the most robust finding in the analysis. In the pooled OLS (Model 1), the energy elasticity is 0.580 (SE = 0.138, $p < 0.001$), rising to 0.722 in the FE baseline (SE = 0.150, $p < 0.001$) and remaining stable across the EKC (0.738, Model 3) and industry-augmented (0.705, Model 4) specifications. H1 is strongly confirmed. A 1% increase in per capita energy use raises per capita CO₂ by approximately 0.72% within GCC countries over time, consistent with Harrathi and Almohaimeed [6] and Balogh and Jámor [4]. The higher FE estimate relative to pooled OLS reflects control for country-level fixed factors (e.g., Qatar’s exceptionally high LNG-driven energy intensity) that are correlated with energy use.

The contrast between the pooled OLS overall R² (0.971) and the FE within-R² (0.513–0.538) reflects a fundamental difference in what each statistic measures, not a contradiction in model fit. The pooled OLS R² captures the proportion of total variance in lnCO₂ explained by the regressors, including the very large between-country differences in emissions levels as follows: Qatar permanently emits approximately twice as much CO₂ per capita as Oman, and this permanent gap—driven by structural differences in hydrocarbon intensity, income, and urban form—is easily fitted by cross-sectional variation in income and energy use. The FE within-R², by contrast, measures only the proportion of within-country, over-time variation in lnCO₂ explained after removing country fixed effects. Within a given GCC country, annual emission changes are smaller, noisier, and partly driven by factors not captured in the model (e.g., year-specific policy announcements, extreme weather events affecting cooling demand). A within-R² of 0.51–0.54 is therefore a credible and economically meaningful goodness-of-fit for within-country emission dynamics over the 2015–2022 regression sample. The appropriate benchmark is the within-R², not the pooled R², for evaluating the FE specifications.

Income and EKC test (H2 and H3). In the pooled OLS (Model 1), lnGDPpc is positive and significant ($\beta = 0.204$, $p = 0.017$), supporting H2. Crucially, the EKC specification (Model 3) reveals a significant inverted-U pattern: the linear lnGDPpc term is strongly positive ($\beta = 8.839$, SE = 2.533, and $p = 0.001$) and the quadratic (lnGDPpc)² term is negative and significant ($\beta = -0.389$, SE = 0.113, and $p = 0.002$), confirming H3. Within

GCC countries, CO₂ per capita first rises, then falls as income increases after controlling for energy use, trade, urbanization, and FDI. The implied EKC income turning point is $\exp(-8.839/(2 \times (-0.389))) \approx \exp(11.36) \approx \text{USD } 85,500$ per capita. This level is already exceeded by Qatar (panel mean USD 114,835) and is approached by the UAE (USD 71,434), consistent with the incipient emission stabilization observed in these two economies during 2015–2024. Lower-income GCC members (Bahrain, Kuwait, Oman, and Saudi Arabia) remain on the EKC's rising slope, consistent with their continued emissions' growth trajectories. These results corroborate Adow et al. [7] and Toukabri and Chaouachi [17].

Trade openness (H4). Trade openness is negative and significant in the pooled OLS ($\beta = -0.080$, $\text{SE} = 0.021$, and $p < 0.001$), supporting H4, and consistent with Harrathi and Almohaimeed's [6] long-run finding of a negative trade–emission relationship for GCC economies. However, the coefficient loses significance in all fixed-effect models. The divergence reflects a Dutch Disease dynamic as follows: in hydrocarbon-exporting economies, the trade/GDP ratio fluctuates procyclically with oil prices, because export revenues and the GDP denominator move together. Year-to-year variation in this ratio, therefore, does not capture genuine shifts in trade integration, and no detectable within-country emissions response is identified over the 2015–2022 window. This is consistent with Aller et al. [3], who report that trade openness is a non-robust determinant of CO₂ in short panels.

Urbanization (H5). Urbanization is positive and significant in FE Models 2 and 4 ($\beta \approx 2.0$, $p < 0.05$), confirming H5 as follows: within GCC countries, a 1% increase in the urban population share is associated with approximately a 2.0% rise in CO₂ per capita. This large elasticity reflects the energy intensity of GCC urban expansion in desert environments—air-conditioning demand, desalination infrastructure, and car-based mobility. The insignificant urbanization coefficient in pooled OLS reflects confounding by cross-country income differences, which the FE estimator resolves by exploiting within-country variation. Results align with Asif et al. [12] and Kwakwa [18].

FDI and industry value added. FDI is statistically insignificant across all specifications, consistent with Harrathi and Almohaimeed [6]. The concentration of GCC FDI in hydrocarbons, real estate, and financial services means that neither the pollution haven nor the pollution halo mechanism dominates. The industry value-added coefficient in Model 4 is negative ($\beta = -0.060$) and insignificant ($p = 0.546$). GCC industry value-added shares fluctuate counter-cyclically with oil prices—when hydrocarbon revenues contract relative to the non-oil economy, the industry's GDP share falls—making it a noisy proxy for physical industrial activity over this horizon.

4.4. Robustness Analysis

Four robustness checks confirm the stability of the core findings. *First*, Model 5 substitutes electric power consumption for total energy use. The coefficient remains positive (0.280) but loses significance ($p = 0.221$), and within-R² falls to 0.274 from 0.513 in Model 2. This confirms that comprehensive energy use—capturing direct fuel combustion in transport and industry beyond electricity generation—is the superior proxy for GCC fossil-fuel intensity. *Second*, the random-effect estimator yields directionally consistent energy ($\beta = 0.400$, $p = 0.025$) and GDP ($\beta = 0.435$, $p = 0.001$) coefficients, confirming the FE and RE estimates broadly agree despite the theoretical preference for FE given GCC countries' systematic structural differences. *Third*, within-R² improves from 0.513 (Model 2) to 0.538 (Model 3), confirming that the quadratic income term adds meaningful explanatory power consistent with an EKC. *Fourth*, the high pooled OLS overall R² (0.971) compared with within-R² (0.513–0.538) reveals that most variance in GCC CO₂ is between countries—captured by fixed effects representing permanent structural differences in hydrocarbon intensity, in-

come levels, and urban form—while within-country dynamics are well characterized by energy use and the income quadratic. Fifth, we re-estimate the baseline fixed-effect model (Model 2) using Prais–Winsten panel-corrected standard errors (PCSEs), which correct for panel-specific first-order autocorrelation and contemporaneous cross-sectional correlation simultaneously. The energy coefficient remains positive and significant, and the qualitative pattern of results is unchanged, further corroborating the robustness of the core findings to alternative variance estimation strategies.

5. Discussion

As noted in Section 3.2, the results are interpreted throughout as conditional associations; causal identification remains an avenue for future research. Policy recommendations are offered as directional guidance consistent with the estimated associations.

5.1. Energy Consumption as the Dominant Driver: Structural Lock-In and Policy Implications

The most consistent and practically relevant result concerns the role of energy consumption in driving CO₂ emissions across GCC economies during 2015–2022. The within-country FE elasticity of 0.72 implies that a 10% reduction in per capita energy use, holding income and urbanization constant, would reduce per capita CO₂ by approximately 7.2%. GCC per capita energy use averages nearly 10,000 kg of oil equivalent annually—more than twice the OECD average—indicating substantial scope for emission reduction through energy restructuring. Oil and gas account for over 90% of GCC electricity generation, and fossil fuel subsidies have increased in recent years despite net-zero pledges [2].

The result aligns closely with Harrathi and Almohaimeed's [6] long-run PMG estimates and Balogh and Jámboř's [4] global GMM evidence. GCC countries have announced substantial renewable energy targets for 2030 [26], with installed renewable capacity rising sharply over the 2015–2022 period [27]. The energy elasticity estimated here suggests that achieving meaningful CO₂ reductions through energy restructuring will require reducing fossil fuel intensity substantially, consistent with these targets.

5.2. EKC Turning Point and the Diverging GCC Emissions Trajectory

The implied EKC income turning point of approximately USD 85,500 per capita (Table 5, Model 3) aligns with recent GCC-specific empirical findings by Adow et al. [7] and Toukabri and Chaouachi [17]. The higher turning point relative to Adow et al. [7] (who find USD 13,867–28,158) reflects the following two design choices: (i) controlling for energy use in the same equation as the income quadratic, which allows income to capture residual technique and composition effects rather than the direct fossil-fuel demand channel; and (ii) the 2015–2022 analytical period, which exhibits stronger emission growth in lower-income GCC members (on the rising EKC slope) alongside incipient stabilization in Qatar and the UAE (at or above the turning point). The FE estimator, by exploiting within-country income variation over the 2015–2022 sample window, captures this heterogeneous dynamic precisely.

The EKC finding has meaningful policy implications beyond the GCC. It suggests the growth–environment trade-off is not permanent: as GCC economies pursue diversification and accumulate the institutional and technological capacity associated with higher incomes, emissions can be expected to decline. However, the literature cautions that EKC dynamics are period-specific and sensitive to governance quality [16]. The technique and composition effects that drive the EKC's downward slope require deliberate policy acceleration through carbon pricing, industrial R&D incentives, and structural diversification.

5.3. Urbanization and the Energy-Intensive Desert City

The significant positive within-country urbanization effect ($\beta \approx 2.0$) requires contextualization. In most global panel studies, urbanization's environmental effect is ambiguous or negative when agglomeration efficiencies dominate [5]. The GCC setting is structurally different: built in desert climates with ambient temperatures frequently exceeding 40 °C, GCC cities require energy-intensive cooling infrastructure. Air conditioning accounts for up to 70% of peak electricity demand [28], virtually all potable water is produced by energy-intensive desalination, and car-dependent urban forms with limited public transport alternatives further amplify per capita energy use. Findings align with Asif et al. [13], who document cointegration between urbanization, energy, and CO₂ for GCC economies, and with Kwakwa [18], who finds robust positive urbanization elasticities for 85% of East African Community countries.

The contrast between the insignificant pooled OLS urbanization coefficient and the significant FE coefficient illustrates the value of the FE identification strategy as follows: cross-sectional variation in urbanization rates is confounded by income levels (higher-income countries are more urbanized), but within-country urbanization growth—purged of permanent structural differences by country fixed effects—reveals a genuine positive emissions impact.

5.4. Trade, FDI, and Industrialization: Cross-Sectional vs. Within-Country Dynamics

The trade and FDI findings carry broader implications beyond what the regression coefficients alone convey. The cross-sectional trade effect—negative and significant in pooled OLS—indicates that more open GCC economies have, over time, developed lower-emission production and consumption structures, consistent with long-run technique-effect gains from market integration. That this relationship disappears in the fixed-effect model does not contradict the cross-sectional finding; it instead reflects the Dutch Disease dynamics discussed in Section 4.3, whereby short-run oil-price-driven variation in the trade/GDP ratio masks the underlying structural channel. Policy implications, therefore, differ by time horizon as follows: in the short run, within-country trade fluctuations cannot be expected to reduce emissions, but structural trade liberalization coupled with investment in clean technology absorption could yield long-run compositional gains. The FDI insignificance is consistent with Aller et al. [3]. The persistence of FDI insignificance across all specifications confirms Harrathi and Almohaimeed [6] and reflects GCC FDI's sectoral concentration in hydrocarbons, real estate, and financial services, where neither the pollution haven nor the halo mechanism dominates. Structural reforms attracting FDI into renewable energy and green technology would be needed to activate the pollution halo effect and transform FDI into an active emission reduction channel.

6. Conclusions

This study has examined the determinants of CO₂ emissions per capita in six GCC economies over 2015–2022, with particular focus on the role of energy consumption, economic growth, and urbanization. Three robust conclusions emerge from pooled OLS and country fixed-effect panel regressions with clustered standard errors.

First, energy use per capita is the dominant driver of CO₂ emissions (within-country FE elasticity ≈ 0.72 , $p < 0.001$), stable across all model specifications and consistent with the GCC's structural fossil-fuel dependence [4,6]. Second, the EKC hypothesis is empirically supported: both the linear ($\beta_2 = 8.839$) and quadratic ($\beta_3 = -0.389$) income terms are significant at the 1% level in the FE framework, with an implied income turning point at approximately USD 85,500 per capita—already exceeded by Qatar and approached by the UAE, consistent with Adow et al. [7] and Toukabri and Chaouachi [17]. Third, urbanization

raises CO₂ significantly within GCC countries ($\beta \approx 2.0, p < 0.05$), reflecting energy-intensive urban expansion in desert environments.

These findings generate three policy implications. *First*, the energy association is the most consistent finding across all specifications, pointing toward energy efficiency improvements and renewable energy substitution as priority policy directions. Subsidy reform, mandatory building and appliance efficiency standards, and accelerated renewable deployment are among the measures most directly aligned with the estimated energy–emission relationship. *Second*, the EKC evidence suggests Qatar and the UAE may be approaching self-sustaining emissions reduction; deliberate policy acceleration—carbon pricing, R&D incentives, and industrial diversification—is needed to realize the EKC’s downward slope without waiting for income effects alone. *Third*, green urban planning from the outset—mandatory building energy codes, district cooling systems, and public transport investment—can lock in low-carbon urban forms in Saudi Arabia and Oman, where urbanization continues. For lower-income GCC members—Oman, Bahrain, Kuwait, and Saudi Arabia—which remain on the rising slope of the EKC, deliberate policy can accelerate the transition toward the turning point without waiting for income effects alone. Two sectors warrant priority. First, utility-scale solar power combined with green hydrogen production leverages the region’s exceptional solar irradiance (among the highest globally) to displace fossil-fuel combustion in both electricity generation and heavy industry. Saudi Arabia’s NEOM green hydrogen project and Oman’s Hyport Duqm initiative represent nascent but scalable models. Second, solar-powered desalination—replacing the current thermally driven multi-stage flash technology with reverse osmosis powered by photovoltaics—can substantially reduce the energy intensity of potable water production, which is a disproportionately large share of GCC electricity demand. Targeted public–private investment in these two sectors would compress the emission trajectory of lower-income GCC members and bring their effective turning point within a shorter time horizon.

Future research should address the following five limitations: extending data through 2025 for all variables, including energy use; expanding the sample to other high-income oil economies; disaggregating emissions by sector; applying dynamic GMM or instrumental variables to address potential endogeneity; and incorporating governance quality indicators to examine institutional mediation of the energy–emission nexus [3,16].

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